VIDUSH RAJ SELVARAJ

AI Engineer | Generative AI & Agentic Systems | Software Engineering

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

AI Engineer with 2+ years of experience building and deploying production-grade Generative AI systems, multi-agent workflows and real-time voice assistants, grounded in a First-Class Honours BEng in Software Engineering and strong backend and full-stack skills (Python, FastAPI, Node.js, Spring Boot, React, Docker, CI/CD). Hands-on across the Google Cloud AI stack (Vertex AI / Gemini Enterprise Agent Platform, ADK, A2A, Gemini Enterprise), the Microsoft AI stack (Microsoft Foundry, Copilot Studio, Power Platform) and Google Workspace automation. Builds and operates MCP servers and custom tools, implements RAG on managed and open-source platforms, and advises companies on AI architecture. Experienced in remote, cross-functional work with international clients.

CORE COMPETENCIES

Agentic AI & Protocols: Google ADK, Agent-to-Agent (A2A), Model Context Protocol (MCP) servers and tools, Vertex AI Agent Builder / Agent Studio, Microsoft Agent Framework, Semantic Kernel, LangChain, LlamaIndex, multi-agent orchestration, function/tool calling, Gemini Live API

RAG & Retrieval: Managed RAG (Vertex AI RAG Engine, Vertex AI Search, Vector Search, Azure AI Search) and open-source RAG (Qdrant, FAISS, Pinecone), Cohere embeddings and reranking, secure RAG pipelines

Model Training & Fine-Tuning: Vertex AI custom training and supervised fine-tuning, Model Garden, Pipelines, Model Registry, Endpoints; TensorFlow, Hugging Face, Transformers, knowledge distillation, NLP

Software Engineering & Backend: Python, Java, TypeScript, Node.js, FastAPI, Flask, Django, Spring Boot, REST and WebSocket APIs, stateless services, JWT / Spring Security, OOP, software testing and QA, Agile

Google Cloud: Vertex AI / Gemini Enterprise Agent Platform, Gemini API, Cloud Run, Cloud Build, Pub/Sub, Cloud Storage, Cloud SQL, BigQuery, Firestore, Secret Manager, Cloud IAM, Speech-to-Text V2, Dialogflow CX, Document AI

Microsoft: Microsoft Foundry (formerly Azure AI Foundry), Foundry Agent Service, Azure OpenAI, Copilot Studio, Microsoft 365 Copilot, Power Automate, Power Apps, Dataverse, Azure Functions, Logic Apps, Microsoft Entra ID

Workspace & Automation: Google Workspace Studio, Gemini Enterprise, n8n, Power Automate, AI-driven process automation, legacy-to-agentic modernisation

Frontend & Mobile: React, Next.js, React Native, Flutter, Dart

DevOps & Data: Docker, GitHub Actions CI/CD, MongoDB, MySQL, vector databases, WebRTC VAD

Consulting & Collaboration: AI solution architecture, requirements gathering, stakeholder communication, cross-functional collaboration, project management

PROFESSIONAL EXPERIENCE

AI Engineer — Finetech Consultancy, Sri Lanka Oct 2025 – Oct 2026

Contract Oct 2025 – Apr 2026 • Full-time Apr 2026 – Oct 2026

- Engineered AXON, a multi-lingual AI voice assistant, with Python, FastAPI and WebSockets, using the Gemini Live API for real-time conversational logic.
- Built Voice Activity Detection (WebRTC VAD) to handle user interruptions, and connected the assistant to PBX networks through MCP for autonomous call routing.
- Deployed scalable, stateless backends on Google Cloud using Cloud Run, Secret Manager, Cloud IAM and Cloud Speech-to-Text V2, with secrets kept in Secret Manager and access controlled through Cloud IAM.
- Designed, built, deployed and managed MCP servers and custom tools that give AI agents secure, structured access to enterprise systems and APIs.
- Built multi-agent systems with Google ADK and the A2A protocol on Vertex AI Agent Builder (now Gemini Enterprise Agent Platform), using Agent Studio and Agent Engine to prototype, deploy and run agents.
- Trained, fine-tuned, evaluated and deployed models on Vertex AI using Model Garden, Vertex AI Studio, Pipelines, Model Registry and Endpoints.
- Designed a secure RAG pipeline with Cohere and Qdrant, plus the Google Maps API for location-based branch recommendations; also implemented managed RAG with Vertex AI RAG Engine, Vertex AI Search, Vector Search and Azure AI Search.

- Built and published AI agents and automated business workflows with Microsoft Copilot Studio and Power Automate, and developed pro-code agents on Microsoft Foundry (formerly Azure AI Foundry) using Foundry Agent Service and Azure OpenAI models.
- Automated cross-app workflows in Google Workspace with Google Workspace Studio and Gemini Enterprise, connecting agents to Gmail, Drive, Sheets and Chat.

Technologies: Google Cloud: Vertex AI / Gemini Enterprise Agent Platform (Agent Builder, Agent Studio, ADK, Agent Engine, Agent Garden, Model Garden, Vertex AI Studio, Pipelines, Model Registry, Endpoints, Evaluation, RAG Engine, Vector Search, Vertex AI Search), Gemini API, Gemini Live API, Gemini Enterprise, Google Workspace Studio, Dialogflow CX, Document AI, Speech-to-Text V2, Cloud Run, Cloud Build, Artifact Registry, Pub/Sub, Cloud Storage, Cloud SQL, BigQuery, Firestore, Secret Manager, Cloud IAM, Cloud Logging and Monitoring. Microsoft: Microsoft Foundry (Azure AI Foundry), Foundry Agent Service, Azure OpenAI, Azure AI Search, Azure AI Speech, Azure AI Document Intelligence, Azure Machine Learning, Microsoft Agent Framework, Semantic Kernel, Copilot Studio, Microsoft 365 Copilot, Power Automate, Power Apps, AI Builder, Dataverse, Azure Functions, Logic Apps, Key Vault, Microsoft Entra ID, Microsoft Graph. Other: MCP, A2A, LangChain, Qdrant, Cohere, Python, FastAPI, WebSockets, WebRTC VAD.

AI Consultant & AI Architect — Freelance, Remote Jul 2024 – Oct 2025

- Consulted for multiple companies on AI architecture: solution design, selecting LLMs and cloud services, and choosing between managed and open-source RAG platforms.
- Designed and deployed secure RAG pipelines using managed and open-source services to streamline complex data extraction for private clients.
- Built bespoke AI automation workflows in n8n to automate data extraction and business processes for private clients.
- Created, deployed and managed MCP servers and tools to connect LLM agents with client data sources and business systems.
- Modernised legacy applications by replacing them with autonomous multi-agent workflows that automate decision-making and task execution, using LangChain and leading LLMs (OpenAI, Gemini).
- Delivered production-ready intelligent agents under strict client confidentiality.

QA Intern — DineTap, Singapore Jul 2023 – Jul 2024

- Executed manual and regression testing for the DineTap app and website; maintained test plans and documentation.
- Worked with developers in Agile sprints to reproduce and resolve critical bugs, tracking issues in ClickUp.
- Trained fellow interns on product functionality, building a quality-driven team culture.

SELECTED PROJECTS

- **CliniGuide** (Flutter, Node.js, Flask, LLM, RAG, FAISS, Knowledge Distillation) — Mobile app for personalized medication recommendations using a custom RAG knowledge base and Chain-of-Thought prompting.
- **Supreme Visualization** (Python, CNN, RNN, LSTM, Tkinter) — Deep learning system that detects anomalies in uploaded CCTV footage, with a custom UI.
- **Skin Consultation Web App** (React, Spring Boot, MongoDB, Spring Security, GCP, Docker) — Booking platform with JWT-secured role-based dashboards, Cloud Storage for images and Cloud Run deployment.
- **Cruise Ship Management App** (React, TypeScript, Spring Boot, Docker, GCP) — Modernised a Java CLI into a full-stack system for bookings, waiting lists and expenses.
- **University Progression Calculator** (React, TypeScript, Flask, Docker, Cloud Run, GitHub Actions) — Web calculator backed by a stateless REST API with automated CI/CD.

EDUCATION

BEng (Hons) Software Engineering — First-Class Honours Sep 2021 – Sep 2025

Informatics Institute of Technology (IIT), Sri Lanka, affiliated with the University of Westminster, UK • Graduated November 2025

LANGUAGES

English • Sinhala • Tamil